

Python file update Algorithm

Project description

In my organization, access to restricted content is managed through an allow list of IP addresses, which is stored in "allow_list.txt." Additionally, a separate remove list specifies IP addresses that should be denied access. I developed an algorithm to automate the updating of the "allow_list.txt" file by removing these specified IP addresses.

```
# Define the file paths
import_file = "allow_list.txt"
remove_list = ["192.168.1.1", "10.0.0.5"] # Example IPs to remove

# Step 1: Open the allow list file and read the file contents into a string
with open(import_file, "r") as file:
    ip_addresses = file.read()

# Step 2: Convert the string to a list
ip_addresses = ip_addresses.split()

# Step 3: Iterate through the remove list and remove IP addresses that are in the remove list
for element in remove_list:
    if element in ip_addresses:
        ip_addresses.remove(element)

# Step 4: Update the file with the revised list
updated_content = "\n".join(ip_addresses)
with open(import_file, "w") as file:
    file.write(updated_content)

print("allow_list.txt updated successfully.")
```

Summary

This algorithm effectively removes specified IP addresses from the "allow_list.txt" file. It involves reading the file, processing the contents into a list, iterating through the remove list to eliminate matches, and updating the file with the revised list.